Oral Surgery
Third stage
Lec. 15

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Mechanical principles of extraction

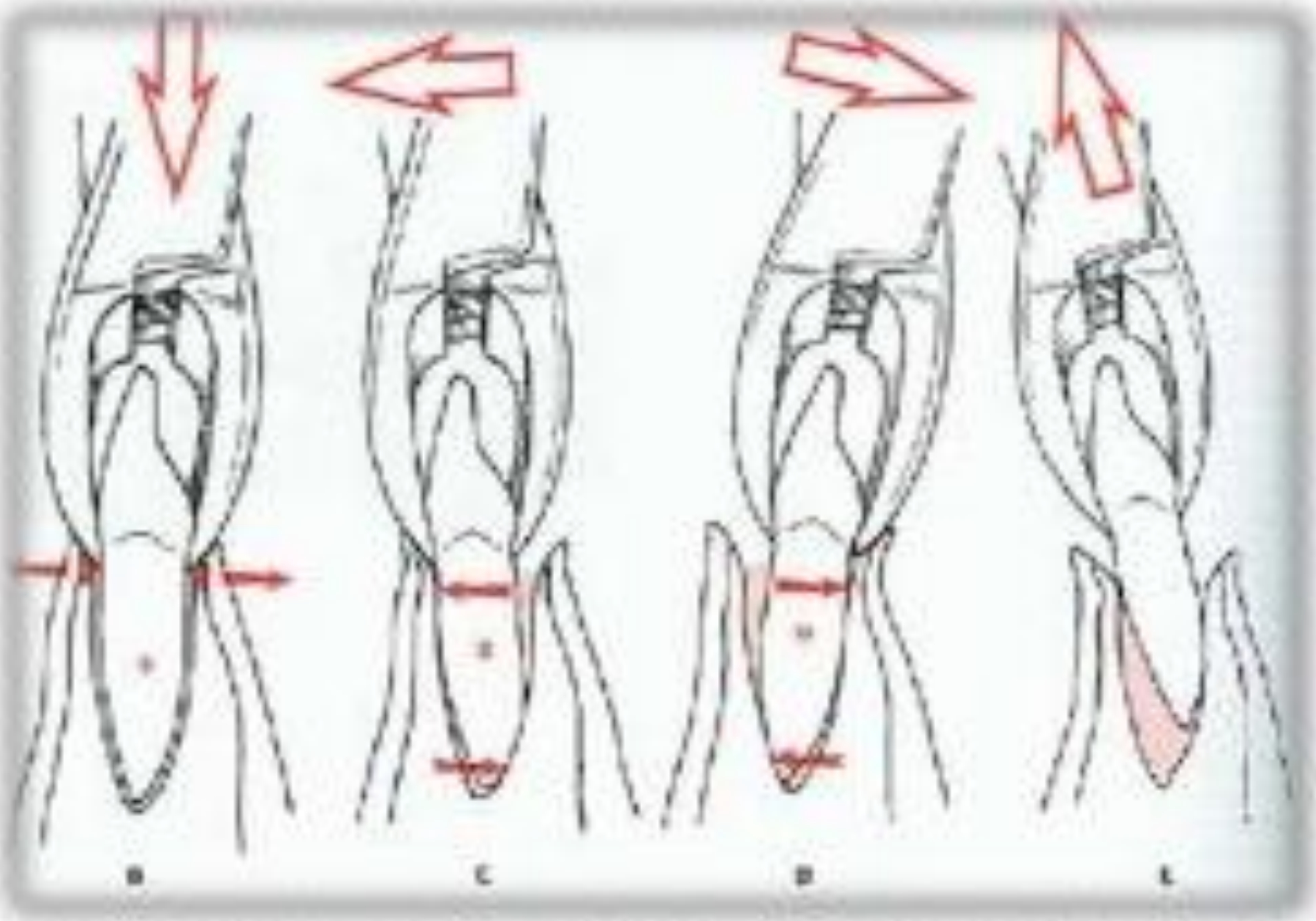
The removal of teeth from the alveolar process employs the use of the following mechanical principles:

I- Expansion of the bony socket:

This is achieved by using the tooth itself as a dilating instrument, and this is the most important factor in forceps extraction, and this principle depend on:

- 1) Sufficient tooth structure be present to be firmly grasped by the forceps.
- 2) The root pattern of the tooth in such that it is possible to dilate the socket to permit the complete dislocation of the tooth from its socket.

- 3)Nature of the bone, elastic bone especially in young patients is maximal and decrease with age, older patients usually have denser, more highly calcified bone that is less likely to provide adequate expansion during extraction of the teeth
- 4) Thickness of the bone: Thick bone expansion is less likely to occur by using normal force.

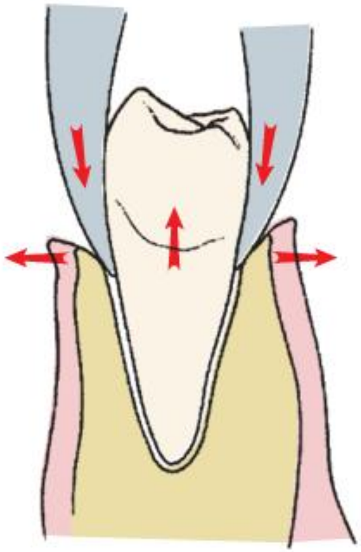


2- The use of a lever and fulcrum:

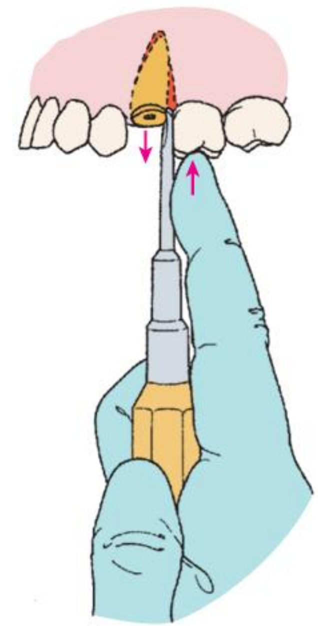
- This is used to force a tooth or root out of the socket along the path of least resistance and this principle is the basic factor governing the using elevators to extract teeth or roots.

3- The insertion of a wedge or wedges:

- Between the tooth-root and the bony socket thus causing the tooth to rise in its socket and this explains why some conically rooted mandibular premolar and molars sometimes shoot out of their socket when forceps blades are applied to it.



Beaks of the forceps act as wedges to expand alveolar bone and displace the tooth in the occlusal direction.



Small, straight elevator used as wedge to displace the tooth root from its socket by driving the elevator apically in the periodontal ligament space.

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Forceps

- Components of a dental extraction forceps consists of:

  - handle

  - hinge

  - beaks

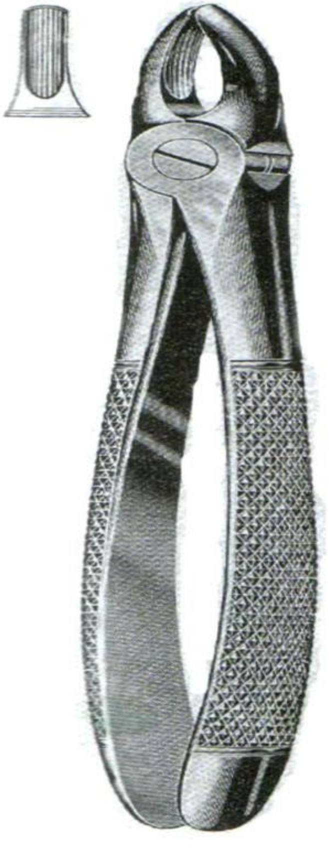
THE FORCEPS FOR UPPER TEETH:

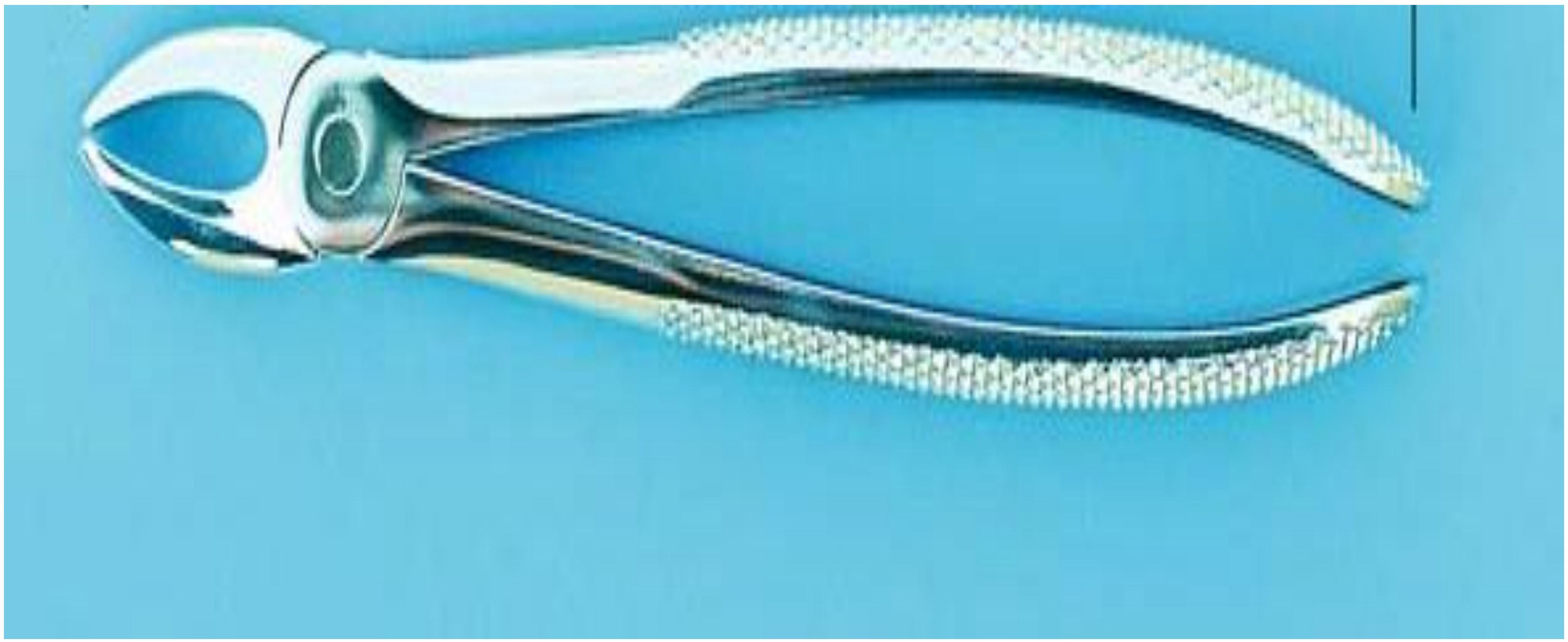
1- The upper straight forceps

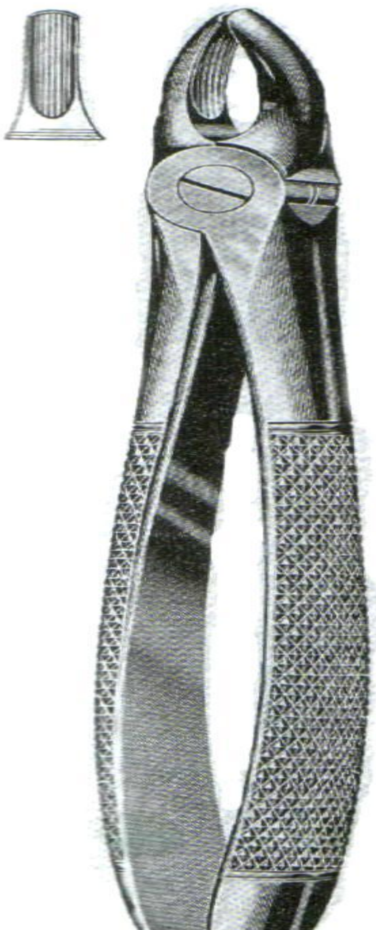
The blades, joint and handle are in one long straight line.

We have two types, one with broad blades that is we called heavy blades and this is used for extraction of upper central incisors and canines, left and right.

The second types of straight forceps has narrow blades or we call it fine blades for extraction of upper lateral incisors left and right and upper anterior retained roots.







2- The upper premolar forceps:

- Here we have two bends in the design of the forceps, these two bends to apply the forceps parallel to long axis of premolar and to avoid injury to the lower lip and apposing teeth.
- The upper premolars teeth has either one root or two roots (one buccal and one palatal), that's mean there is no difference in the anatomy of the tooth root of the premolar on the buccal and palatal surface so the two blades of the premolars forceps are mirror image to each other.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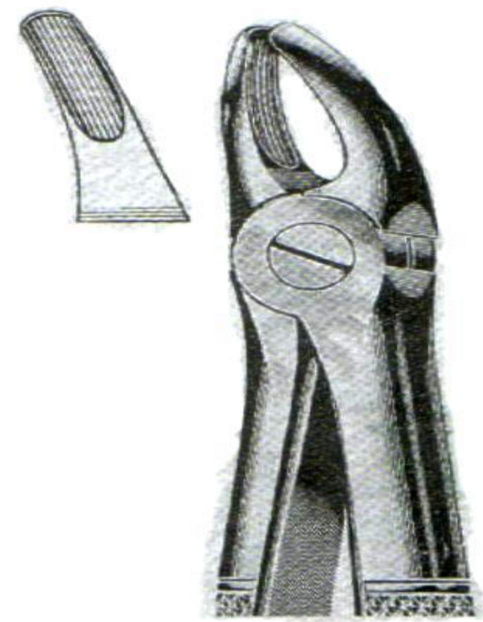


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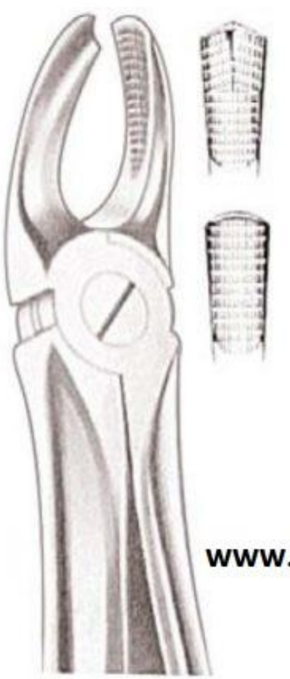
3- The upper molar forceps (full crown upper molar forceps)

- Since upper molar teeth have three roots, two buccal and one palatal, the blade of palatal side is round to fit on palatal root, while blades on buccal has pointed tip or projection so it can enter or fit the bifurcation between the two buccal roots (mesial and distal on the buccal side of the tooth.
- So we have two forceps one for the right molars and one for the left molars and these forceps also double bend for the same requirement as mentioned for premolar teeth



S26.07
Upper Molars Right
Fig. 17

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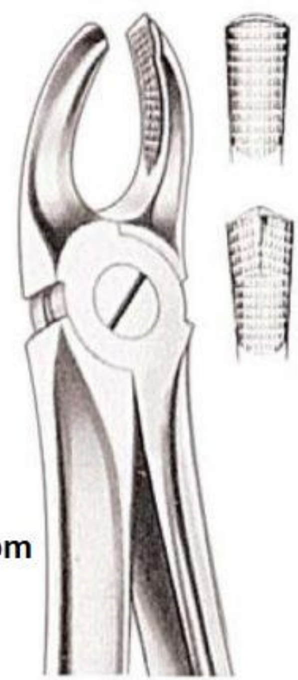

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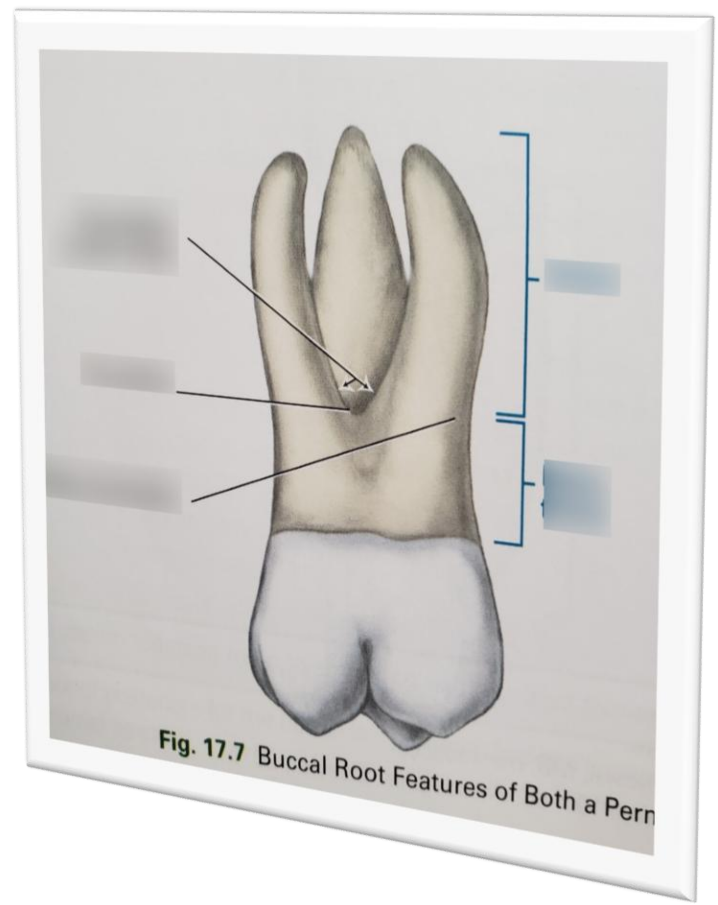
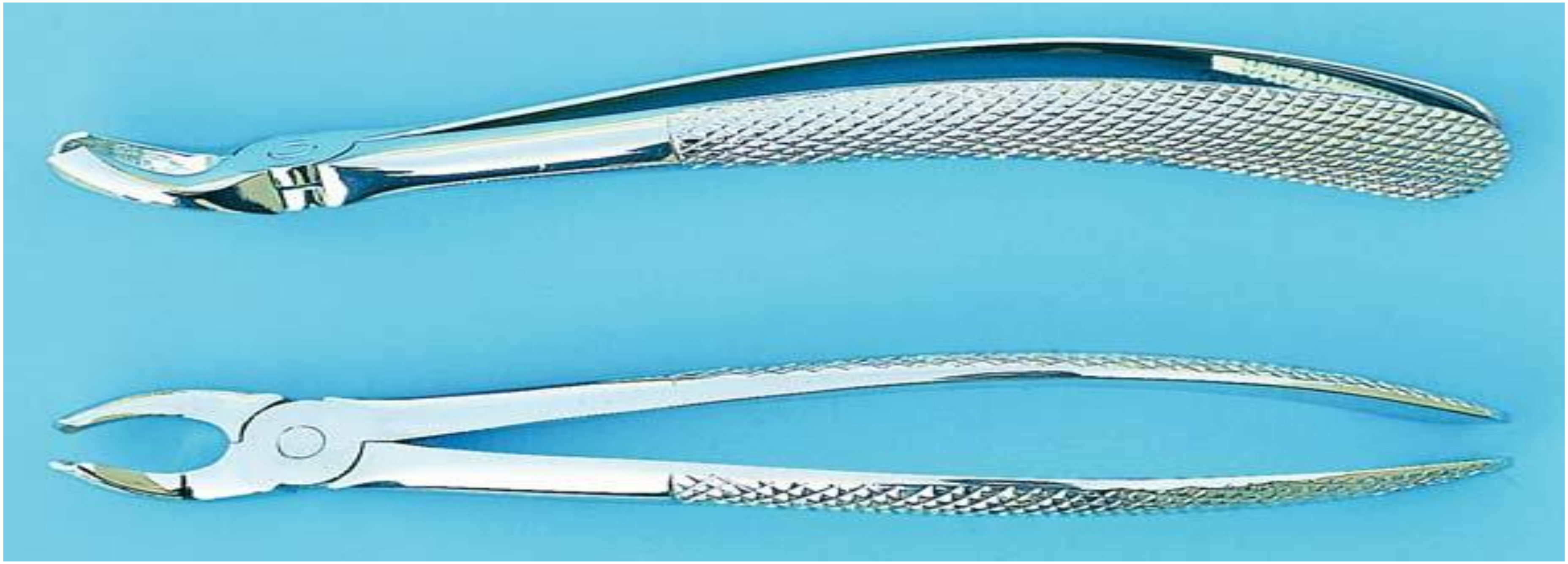
Upper Molars Left
Fig. 18





- The Bayonet forceps, the blades of the forceps offset so the long axis of the blades parallel to long axis of the handle.
- This forceps is suitable for extraction of upper 3 molars right and left.
- In addition of these forceps there are two forceps for extraction of retained roots, there are fine blades, one forceps is straight for upper anterior retained roots and one curved for upper posterior roots.

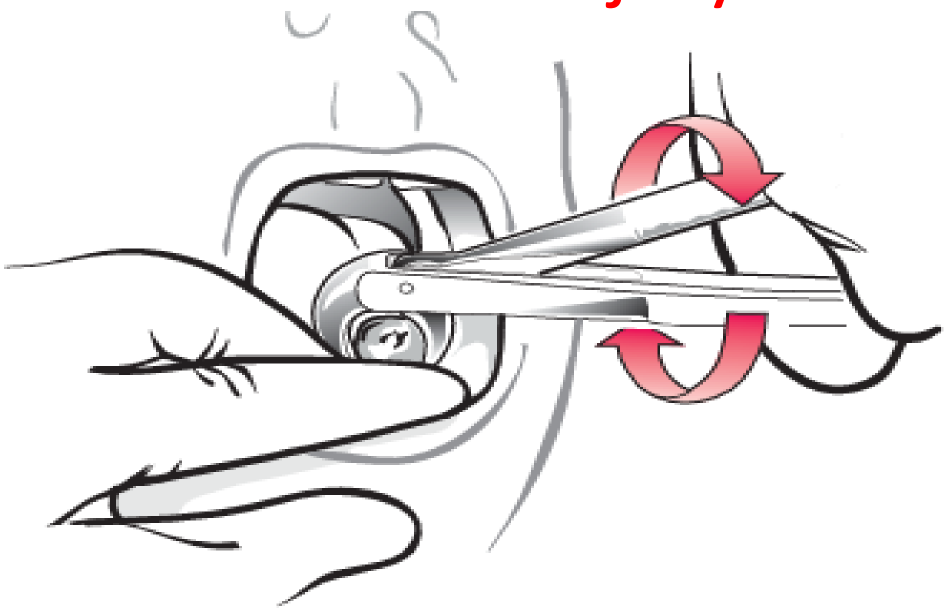
Bayonet forceps

Bayonet forceps

Upper anterior retained root **Upper posterior retained root**

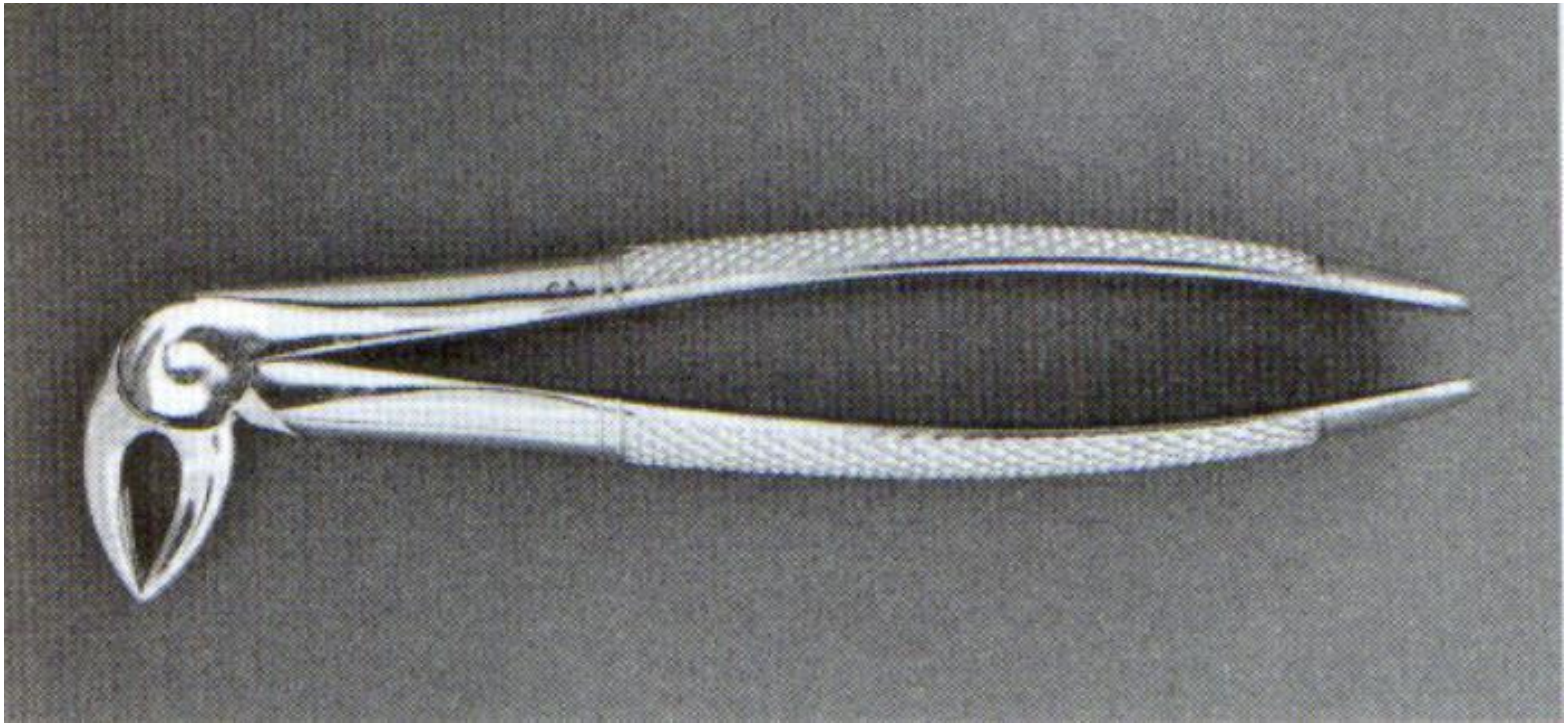
THE FORCEPS OF LOWER TEETH:

- Here we have the long axis of the blades is in right angle to the long axis of the handle so the blades can be applied apical to the cemento-enamel junction (on the root of the tooth surface parallel to the long axis of the tooth and the handle not to cause injury to the upper jaw.



1- Forceps for extraction of lower anterior teeth

- We have fine blades for extraction of the lower central and lateral incisors which have fine roots with flattened sides mesiodistally and heavy blades used for extraction of canines.



2- Premolar forceps:-

- Because the bucco-lingual width of the crown in the premolar teeth is larger than that of lower incisors and canines we use forceps with heavy blades but partially away from each other when close to accommodate the crowns of these teeth without crushing for the crown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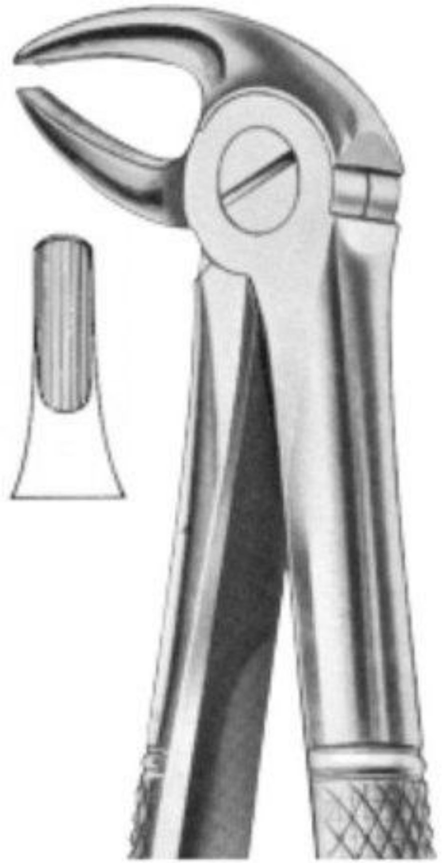
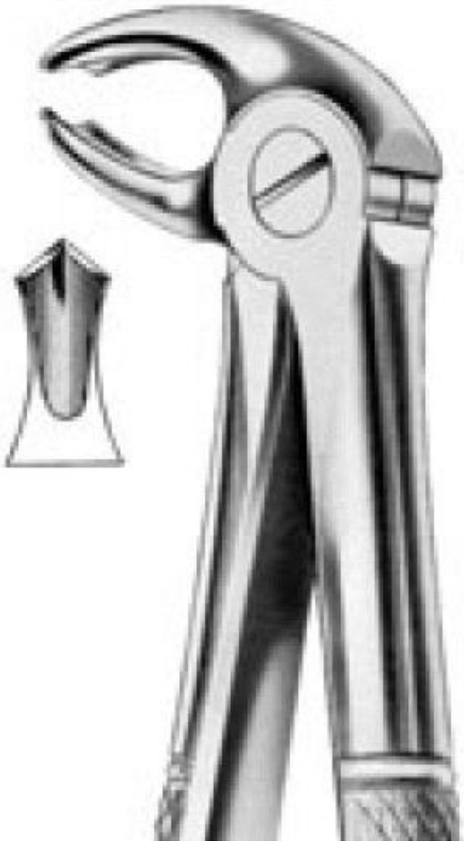
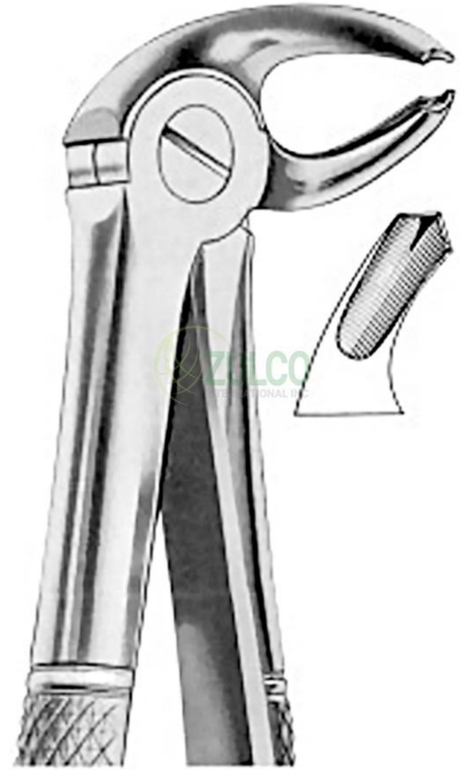


3- Full crown lower molar forceps

Since the lower molar teeth has two roots, one mesial and one distal root so the buccal and lingual blades of the forceps designed with projected tapered tip to fit the bifurcation of these teeth on the buccal and lingual sides, so the buccal and lingual blades are identical so the same forceps can be used on the right and left sides on opposite to that in upper molar teeth.

In addition to that we have a fine blades forceps for extraction of retained roots anteriorly and posteriorly also we have two Bayonet forceps for lower 3rd molars; one for left side and the other for right side.





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Lower Molars

Thank You